

Image Recognition Pipeline

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```
# Disable warnings and messages globally during render
knitr::opts_chunk$set(
  warning = FALSE,
  message = FALSE,
  verbose = FALSE,
  comment = NA
)
```

```
# Environment setup
Sys.setenv(TF_CPP_MIN_LOG_LEVEL = "3")
Sys.setenv(TF_ENABLE_ONEDNN_OPTS = "0")

library(EBImage)
library(keras3)
```

```
##
## Attaching package: 'keras3'

## The following object is masked from 'package:EBImage':
##
##      normalize
```

```
img_dir <- "C:/SEM VII/R/imgs"
pictures <- c('p1.jpg', 'p2.jpg', 'p3.jpg', 'p4.jpg', 'p5.jpg', 'p6.jpg',
              'c1.jpg', 'c2.jpg', 'c3.jpg', 'c4.jpg', 'c5.jpg', 'c6.jpg')

picarr <- list()
for(i in 1:12) {
  picarr[[i]] <- readImage(file.path(img_dir, pictures[i]))
}

# Preprocessing
for(i in 1:12) {
  picarr[[i]] <- resize(picarr[[i]], 28, 28)
}

for (i in 1:12) {
  picarr[[i]] <- array_reshape(picarr[[i]], c(28, 28, 3))
}

# Dataset split
```

```

x_train <- NULL
for (i in c(1:5, 7:11)) {
  x_train <- rbind(x_train, as.vector(picarr[[i]]))
}

# Vectorize test images as well to match x_train dimensions
x_test <- rbind(as.vector(picarr[[6]]), as.vector(picarr[[12]]))

y_train <- c(0,0,0,0,0,1,1,1,1,1)
y_test  <- c(0, 1)

train_labels <- to_categorical(y_train)
test_labels  <- to_categorical(y_test)

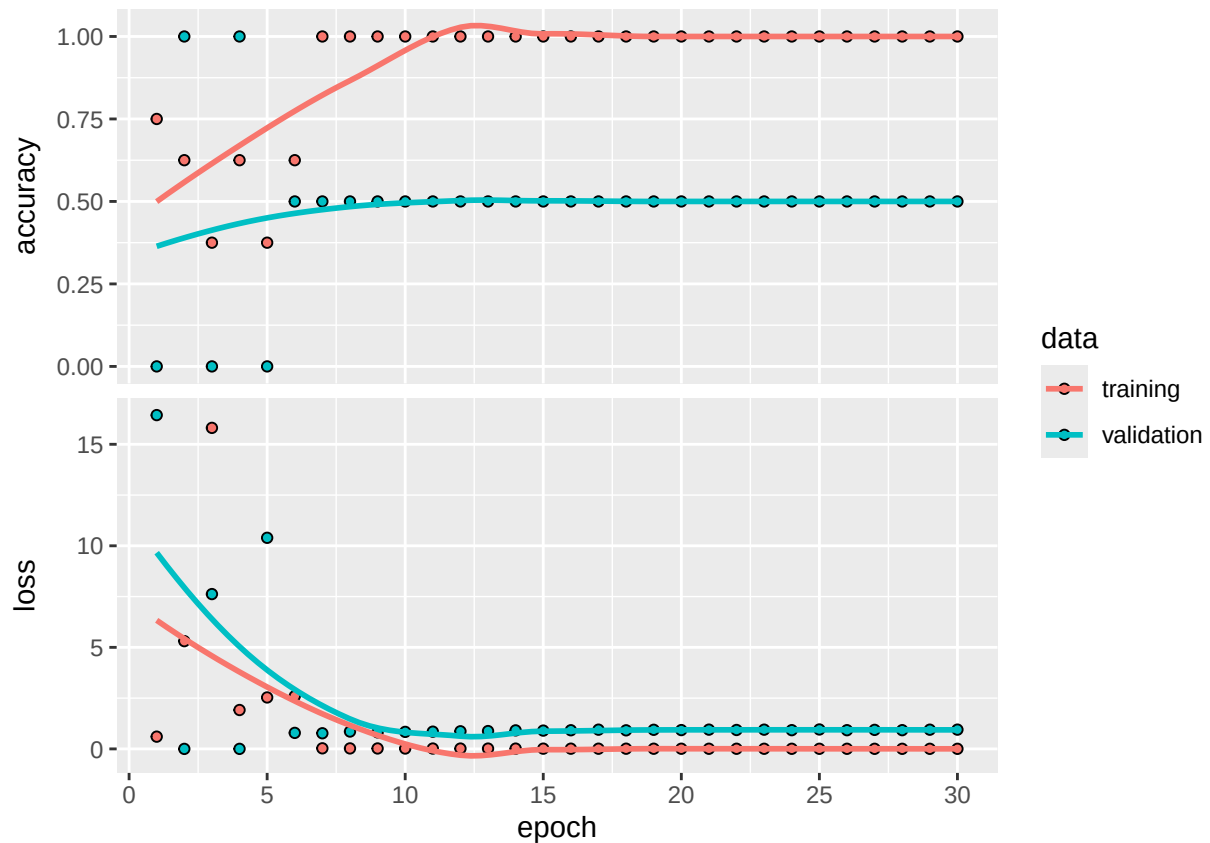
# Model architecture
model <- keras_model_sequential(
  layers = list(
    keras_input(shape = c(2352)),
    layer_dense(units = 256, activation = 'relu'),
    layer_dense(units = 128, activation = 'relu'),
    layer_dense(units = 2, activation = 'softmax')
  )
)

model %>% compile(
  loss = 'categorical_crossentropy', # Changed from binary_crossentropy to match 2-unit softmax output
  optimizer = optimizer_rmsprop(),
  metrics = c('accuracy')
)

# verbose = 0 disables the unicode progress bar that breaks pdflatex
history <- model %>% fit(
  x_train, train_labels,
  epochs = 30,
  batch_size = 32,
  validation_split = 0.2,
  verbose = 0
)

plot(history)

```



```
# Evaluation & Predictions
```

```
model %>% evaluate(x_train, train_labels, verbose = 0)
```

```
## $accuracy
## [1] 0.9
##
## $loss
## [1] 0.1927542
```

```
prob <- model %>% predict(x_train, verbose = 0)
pred <- max.col(prob) - 1
table(Predicted = pred, Actual = y_train)
```

```
##           Actual
## Predicted 0 1
##           0 5 1
##           1 0 4
```

```
cbind(prob, Predicted = pred, Actual = y_train)
```

```
##
##           Predicted Actual
## [1,] 0.997206867 0.002793101      0      0
## [2,] 0.998553097 0.001446914      0      0
## [3,] 0.998494267 0.001505680      0      0
```

##	[4,]	0.995240450	0.004759575	0	0
##	[5,]	0.996096909	0.003903082	0	0
##	[6,]	0.010477962	0.989522040	1	1
##	[7,]	0.001429592	0.998570442	1	1
##	[8,]	0.003738060	0.996261895	1	1
##	[9,]	0.006660808	0.993339121	1	1
##	[10,]	0.849036336	0.150963634	0	1